

A FOUR-STRATUM FUNCTIONAL GRAPH AND EXACT FIBRES FOR A BINARY-PROJECTIVE STEINER TRIANGLE MAP

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ABSTRACT. Let $\mathcal{P} = \mathbb{F}_2^r \setminus \{0\}$ and endow it with the standard Steiner quasigroup operation $x \star x = x$ and $x \star y = x + y$ for $x \neq y$. We determine the complete functional graph of the ordered-triple map

$$(a, b, c) \mapsto (b \star c, c \star a, a \star b).$$

Diagonal triples and projective-line blocks are fixed, triples with exactly two equal entries form strict 3-cycles, and every remaining triple has depth one. If $N = 2^r - 1$, the fixed, strict-3-cycle, and depth-one counts are N^2 , $N(N - 1)$, and $N(N - 1)(N - 3)$, respectively. More sharply, every target fibre has size 1, $N - 2$, or 0 according as the target is diagonal or two-equal, a distinct block, or a nonblock; block sources admit an explicit one-parameter description. We also identify the symmetry orbits and weak components, compute all fibre moments, and recover r from the unlabelled fibre multiset. The Steiner system, its quasigroup operation, and characteristic-two projective model are established inputs and receive no contribution credit. An exact audit checks the claimed graph and inverse formulas through rank six.

1. OWNED INPUTS AND THE FINITE MAP

Fix $r \geq 2$, put $V = \mathbb{F}_2^r$, and let

$$\mathcal{P} = V \setminus \{0\}, \quad N = |\mathcal{P}| = 2^r - 1.$$

The points and lines of $\text{PG}(r - 1, 2)$ form the classical projective Steiner triple system: the line through distinct $x, y \in \mathcal{P}$ is $\{x, y, x + y\}$ [2, 3]. The associated Steiner quasigroup is

$$(1) \quad x \star y = \begin{cases} x, & x = y, \\ x + y, & x \neq y. \end{cases}$$

The correspondence between Steiner triple systems and Steiner quasigroups is standard [1]. Quasigroup and Steiner local rules also occur in cellular automata [5]. All of these facts are background here.

We study the finite self-map

$$(2) \quad S : \mathcal{P}^3 \longrightarrow \mathcal{P}^3, \quad S(a, b, c) = (b \star c, c \star a, a \star b).$$

The recent ‘‘Steiner products and dynamics’’ of Kettinger–Peterson [4] concern an orientation-dependent anticommutative bilinear product on a real vector space and its fixed-vector iterates. That is not the coordinate map (2). This title-level neighbour is included to make the source boundary explicit, not to convert a bounded search non-hit into a priority claim.

TABLE 1. Source subtraction. The middle column is established input; the last column is the residual analyzed here.

Source area	Zero-credit input	Retained special-map output
Steiner systems and quasigroups	operation (1), projective point-line blocks, standard identities	no carrier or quasigroup claim
Quasigroup cellular automata	algebraic local rules and their predictability	finite ordered-triple graph, not a lattice CA
Oriented Steiner products	a real bilinear product and fixed-vector iteration	coordinatewise map (2), four strata, and target fibres

Call a triple *diagonal* if all entries agree, *two-equal* if exactly two agree, a *block* if its entries are distinct and $a + b + c = 0$, and a *nonblock* otherwise. These four classes partition $\mathcal{P}^3$.

Theorem 1 (complete graph and target fibres). *For the map S in (2) the following statements hold.*

- (i) *Every diagonal and every block is fixed. Every two-equal triple has exact period three. Every nonblock has depth one and maps to a fixed block.*

(ii) There are N^2 fixed points, $N(N-1)$ strict 3-cycles, and $N(N-1)(N-3)$ depth-one states. Hence

$$(3) \quad |\text{im } S| = |\text{Per } S| = 4N^2 - 3N.$$

The maximum tail is zero for $r = 2$ and one for $r \geq 3$.

(iii) For every target $q \in \mathcal{P}^3$,

$$(4) \quad |S^{-1}(q)| = \begin{cases} 1, & q \text{ is diagonal or two-equal,} \\ N-2, & q \text{ is a block,} \\ 0, & q \text{ is a nonblock.} \end{cases}$$

For a block $q = (x, y, z)$ its sources are exactly

$$(5) \quad (a, b, c) = (t, t+z, t+y), \quad t \in V \setminus \{0, y, z\}.$$

The same fibre formula holds for S^k for every $k \geq 1$.

(iv) The fixed-point counts and Artin–Mazur zeta function are

$$(6) \quad |\text{Fix}(S^k)| = \begin{cases} N^2, & 3 \nmid k, \\ 4N^2 - 3N, & 3 \mid k, \end{cases} \quad \zeta_S(q) = \frac{1}{(1-q)^{N^2}(1-q^3)^{N(N-1)}}.$$

Proposition 2 (symmetry, components, and fibre moments). *The map is equivariant for the natural action of $\text{GL}(r, 2) \times \mathfrak{S}_3$. For $r \geq 3$, the four classes above are exactly its four orbits on $\mathcal{P}^3$; for $r = 2$ the nonblock orbit is empty. The functional graph has*

$$(7) \quad 2N^2 - N$$

weak components: N isolated diagonal fixed points, $N(N-1)$ isolated 3-cycles, and $N(N-1)$ rooted stars whose centers are block fixed points and whose $N-3$ leaves are nonblocks. In particular, for every integer $j \geq 1$,

$$(8) \quad \sum_{q \in \mathcal{P}^3} |S^{-1}(q)|^j = 3N^2 - 2N + N(N-1)(N-2)^j.$$

For $N > 3$ the maximum fibre is $N-2$ and is attained exactly at blocks; the proportion of Garden-of-Eden states is $(N-1)(N-3)/N^2$, which tends to one with r .

Corollary 3 (fibre enumerator and rank recovery). *The codomain fibre enumerator is*

$$(9) \quad \Phi_r(u) := \sum_{q \in \mathcal{P}^3} u^{|S^{-1}(q)|} = N(N-1)(N-3) + (3N^2 - 2N)u + N(N-1)u^{N-2}.$$

Terms with equal exponents are understood to combine; in particular, when $r = 2$ every target has fibre one. For every $r \geq 2$, the unlabelled fibre multiset alone recovers the projective rank:

$$(10) \quad r = \log_2(\max_q |S^{-1}(q)| + 3).$$

For a uniformly chosen codomain target, the probability of being in the image is $4/N - 3/N^2$, whereas the Garden-of-Eden probability tends to one.

The retained result is the conjunction of the four-stratum graph and the target-resolved inverse law. Neither the projective design nor the operation (1) is claimed.

For orientation, Table 2 displays consequences at the first three ranks. Rank two has no transient shell; rank three is the first case in which most states have no predecessor.

TABLE 2. Exact consequences of Theorem 1. The “3-cycles” column counts cycles rather than states.

r	N	states	fixed	3-cycles	depth one
2	3	27	9	6	0
3	7	343	49	42	168
4	15	3375	225	210	2520

2. THE FOUR STRATA

Proof of Theorem 1(i). Idempotence makes every (x, x, x) fixed. If $x \neq y$, direct use of (1) gives, for one of the three possible equality positions,

$$(11) \quad (x, x, y) \mapsto (x + y, x + y, x) \mapsto (y, y, x + y) \mapsto (x, x, y).$$

The three displayed states are distinct, so the period is exactly three. The other equality positions behave by the same cyclic coordinate symmetry.

For pairwise distinct a, b, c , no idempotent branch is used, and

$$(12) \quad S(a, b, c) = (b + c, c + a, a + b).$$

These three outputs are nonzero and pairwise distinct: a zero output or an equality would contradict distinctness of the inputs. Their sum is zero. Thus every distinct triple maps to a block. If $a + b + c = 0$, then $b + c = a$, $c + a = b$, and $a + b = c$, so the block is fixed. A nonblock cannot be fixed and therefore has depth exactly one. $\square$

Proof of Theorem 1(ii). There are N diagonals. An ordered block is determined by an ordered pair of distinct first entries, because the third is their nonzero sum. Hence there are $N(N - 1)$ blocks and N^2 fixed points in total.

There are $3N(N - 1)$ two-equal states. Equation (11) partitions them into $N(N - 1)$ strict cycles. The number left after removing the diagonal, two-equal, and block strata from N^3 is

$$N^3 - N - 3N(N - 1) - N(N - 1) = N(N - 1)(N - 3).$$

These are precisely the depth-one nonblocks. The image contains the entire periodic locus. It contains nothing else by the classification, proving (3). The depth-one count vanishes exactly when $N = 3$, which is the boundary rank $r = 2$. $\square$

The boundary is worth emphasizing. On the three-point projective line every distinct ordered triple is already a block, so small-rank inspection alone would hide the collapsing shell. The first Garden-of-Eden states occur in the Fano carrier $r = 3$.

3. A DIRECT INVERSE PARAMETRIZATION

Proof of Theorem 1(iii). The equality calculation in (11) shows directly that each diagonal or two-equal target has its unique predecessor in the same stratum. Equation (12) shows that a distinct source can reach only a block; hence a nonblock target has no source.

It remains to invert a block (x, y, z) . Any distinct source must solve

$$(13) \quad b + c = x, \quad c + a = y, \quad a + b = z.$$

The block condition $x + y + z = 0$ is exactly the consistency relation. Set $a = t$. Then (13) gives the family (5). Its entries are all nonzero exactly when $t \notin \{0, z, y\}$. They are then automatically distinct: for example $t = t + z$ would force $z = 0$, and the other two equalities similarly force $y = 0$ or $x = y + z = 0$. Conversely every allowed t yields a valid source. Since $|V| = N + 1$, there are $N - 2$ sources.

Among them, $t = x$ recovers the fixed block itself; the remaining $N - 3$ are nonblocks and have no predecessors. Thus passing from S to any positive iterate neither gains nor loses a source for a block. The same is immediate on the two permutation strata, proving the last assertion. $\square$

The inverse proof is independent of division by the orbit census: it solves the target equations and exposes the three forbidden parameter values.

Proof of Proposition 2. An invertible linear map preserves equality and sends $x + y$ to the sum of the images. A coordinate permutation merely permutes the three opposite pair products. These observations prove equivariance. Diagonals are one linear orbit. A two-equal triple is an ordered independent pair together with an equality position, blocks are ordered independent pairs completed by their sum, and nonblocks are ordered independent triples. Transitivity of $\text{GL}(r, 2)$ on ordered independent lists, together with the coordinate action, gives the orbit claim.

Theorem 1 shows that a diagonal has no attached transient, a 3-cycle has no attached transient, and each block has exactly $N - 3$ nonblock sources in addition to itself. This proves (7) and the component description. There are $N + 3N(N - 1) = 3N^2 - 2N$ targets of fibre one and $N(N - 1)$ block targets of fibre $N - 2$; all others have fibre zero. This gives (8). The remaining assertions follow from the nonblock count in Theorem 1(ii). $\square$

Proof of Corollary 3. Insert the three target counts used in the proof of Proposition 2 into the definition of Φ_r . When Theorem 1(iii) gives maximum fibre $N - 2$; hence $N = \max |S^{-1}(q)| + 2$ and $2^r = N + 1$, proving (10). Finally divide (3) by N^3 . Its complement is the nonblock ratio already stated in Proposition 2. $\square$

Proof of Theorem 1(iv). Only fixed points and strict 3-cycles are recurrent. Therefore a recurrent state is fixed by S^k precisely as stated in (6). Substitution in the defining exponential identity $\zeta_S(q) = \exp(\sum_{k \geq 1} |\text{Fix}(S^k)| q^k / k)$, or equivalently multiplication over primitive cycles, gives the displayed product. $\square$

4. EXACT AUDIT AND LIMITATIONS

The accompanying standard-library verifier enumerates every state through $r = 6$. For each rank it compares the literal update with the four-way classification, orbit depth and period, every target indegree, the explicit parameter set (5), all counts, and fixed points of several iterates. This is finite falsification pressure and regression control, not an all-rank proof or an ownership certificate.

The result is specific to the binary projective Steiner quasigroup and to ordered triples with the simultaneous pairwise update (2). It does not classify arbitrary Steiner quasigroups or triple systems, assert stability under perturbing the rule, or furnish a general quasigroup-dynamics theorem. The source search was bounded and terminology dependent; no novelty, priority, or external-release claim is made.

Data availability. No empirical data were used. The exact verifier and frozen transcript are included with the manuscript.

Ethics, contributions, conflicts, and funding. No human or animal subjects are involved. Authorship is anonymous for internal review. The authors declare no conflict of interest and no external funding for this work.

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